# Exercise Sheet 18

Throughout, $H$ is the Heisenberg group of real matrices

$$(a,b,c) = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

with group law

$$(a,b,c)(a',b',c') = (a+a', b+b', c+c'+ab').$$

The basis of $\mathfrak{h}$ is $X = E_{12}$, $Y = E_{23}$, $Z = E_{13}$. For $\lambda \in \mathbb{R}$ the Schrödinger representation $\Pi_\lambda$ on $L^2(\mathbb{R})$ is given by

$$(\Pi_\lambda(a,b,c)f)(t) = e^{i\lambda(c+bt)} f(t+a).$$

The inner product is $\langle f, h\rangle = \int \overline{f} h \, dt$. Each exercise restates any further data it uses.

**Exercise 18.1 (Group law).**

(a) Derive the group law from matrix multiplication. For $g = (1,2,3)$ and $g' = (2,-1,5)$ compute $gg'$, $g'g$ and $g^{-1}$. Check $gg' = (0,0,ab'-a'b)g'g$.

(b) Show that $\exp(xX + yY + zZ) = (x, y, z + \frac{1}{2}xy)$. Find $W \in \mathfrak{h}$ with $\exp W = (2,3,1)$.

(c) Using $\mathrm{Ad}(g)W = gWg^{-1}$, compute $\mathrm{Ad}(g)$ on $X, Y, Z$ for $g = (a,b,c)$. Write its matrix in the basis $(X,Y,Z)$ and check $\mathrm{Ad}(g)[X,Y] = [\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y]$.

(d) Show that Lebesgue measure $da\,db\,dc$ is invariant under left and right translations.

**Exercise 18.2 (Derivatives and commutation).** Fix $\lambda \in \mathbb{R}$ and let $\mathcal{D} = C_c^\infty(\mathbb{R})$. For $W = xX + yY + zZ$ and $f \in \mathcal{D}$, you may assume that $s^{-1}(\Pi_\lambda(\exp sW)f - f)$ converges in $L^2(\mathbb{R})$ to its pointwise $s$-derivative at $s = 0$. Call this limit $d\Pi_\lambda(W)f$.

(a) Show that $d\Pi_\lambda(W)f = xf' + i\lambda(yt + z)f$.

(b) Check that $[d\Pi_\lambda(W_1), d\Pi_\lambda(W_2)] = d\Pi_\lambda([W_1, W_2])$ on $\mathcal{D}$.

(c) Show that $i\,d\Pi_\lambda(W)$ is symmetric on $\mathcal{D}$.

(d) For $g = (a,b,0)$ and $g' = (a',b',0)$, compute $\Pi_\lambda(g)\Pi_\lambda(g')$ and $\Pi_\lambda(g')\Pi_\lambda(g)$ directly. Show that they differ by the factor $e^{i\lambda(ab'-a'b)}$.

**Exercise 18.3 (No bounded canonical pair).** Let $\mathcal{H} \neq 0$ be a Hilbert space, let $\hbar > 0$, and let $P, Q$ be bounded operators on $\mathcal{H}$ with $QP - PQ = i\hbar I$.

(a) If $\dim \mathcal{H} < \infty$, derive a contradiction using the trace.

(b) Show by induction that $Q^n P - PQ^n = i\hbar\, nQ^{n-1}$ for $n \geq 1$.

(c) Show that $Q^n \neq 0$ for every $n \geq 0$.

(d) Derive a contradiction in general. Explain why the lecture uses the Weyl relations instead.

**Exercise 18.4 (Finite-dimensional representations).** Let $\rho\colon H \to \mathrm{U}(n)$ be a continuous homomorphism with $n \geq 1$. You may use that every continuous homomorphism $\mathbb{R} \to \mathrm{U}(1)$ has the form $c \mapsto e^{i\lambda c}$ with $\lambda \in \mathbb{R}$.

(a) Suppose $\rho$ is irreducible. Show that $\rho(0,0,c) = e^{i\lambda c} I$ for some $\lambda \in \mathbb{R}$.

(b) Suppose $\rho(0,0,c) = e^{i\lambda c} I$ for all $c$. Show that $\lambda = 0$.

(c) Deduce that every irreducible $\rho$ equals some $\chi_{\alpha,\beta}(a,b,c) = e^{i(\alpha a + \beta b)}$.

(d) Show that for $\lambda \neq 0$, $\Pi_\lambda$ has no nonzero finite-dimensional invariant subspace.

**Exercise 18.5 (The induced model).** Fix $\lambda \in \mathbb{R}$. Let $A = \{(0,b,c)\}$ and $\chi(0,b,c) = e^{i\lambda c}$. Let $\mathcal{F}$ be the space of continuous $F\colon H \to \mathbb{C}$ with $F(xn) = \chi(n)^{-1} F(x)$ for $n \in A$, such that $t \mapsto F(t,0,0)$ has compact support. Let $H$ act by $(L_g F)(x) = F(g^{-1}x)$. Give $\mathcal{F}$ the norm $\|F\|^2 = \int |F(t,0,0)|^2 dt$.

(a) Show that $A$ is a normal subgroup and $\chi$ is a character. Show that $(a,b,c) = (a,0,0)(0,b,c-ab)$, and that this factorisation is unique.

(b) Show that $L_g$ preserves $\mathcal{F}$. With $f(t) = F(t,0,0)$, show that $(L_g F)(t,0,0) = e^{i\lambda(c-bt)} f(t-a)$ for $g = (a,b,c)$.

(c) Let $(\mathrm{Ind}(g)f)(t) = e^{i\lambda(c-bt)} f(t-a)$ on $L^2(\mathbb{R})$, and $(Rf)(t) = f(-t)$. Show that $R\,\mathrm{Ind}(g)R^{-1} = \Pi_\lambda(g)$.

**Exercise 18.6 (Automorphisms and inequivalence).**

(a) For $s \neq 0$ let $\delta_s(a,b,c) = (a, sb, sc)$. Show that $\delta_s$ is an automorphism of $H$ and that $\Pi_\lambda \circ \delta_s = \Pi_{s\lambda}$.

(b) Assume $\Pi_1$ is irreducible. Deduce that $\Pi_\lambda$ is irreducible for every $\lambda \neq 0$.

(c) Show that $\Pi_\lambda$ and $\Pi_\mu$ are inequivalent when $\lambda \neq \mu$.

(d) For $r > 0$ let $(D_r f)(t) = r^{1/2} f(rt)$. Show that $D_r$ is unitary and $D_r \Pi_\lambda(a,b,c) D_r^{-1} = \Pi_\lambda(a/r, rb, c)$. Show that $\alpha_r(a,b,c) = (a/r, rb, c)$ is an automorphism. Explain how this fits the Stone–von Neumann theorem.

**Exercise 18.7 (Optional extension: von Neumann's operator).** Fix $\lambda \neq 0$, let $\varphi(t) = \pi^{-1/4} e^{-t^2/2}$, and put $g_{a,b} = \exp(aX + bY) = (a, b, ab/2)$. Define

$$B(f,h) = \frac{|\lambda|}{2\pi} \int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \, \langle f, \Pi_\lambda(g_{a,b}) h \rangle \, da\, db.$$

(a) Show that $|B(f,h)| \leq 2\|f\|\|h\|$. Deduce that $B(f,h) = \langle f, Eh \rangle$ for a bounded operator $E$ with $\|E\| \leq 2$.

(b) Using the lecture's formula $\langle \varphi, \Pi_\lambda(a,b,c)\varphi \rangle = e^{i\lambda(c-ab/2)} e^{-(a^2+\lambda^2 b^2)/4}$, show that $\langle \varphi, E\varphi \rangle = 1$.

(c) Show that $E$ is self-adjoint.